

The Real Time Database on the ECB Data Portal

Experience Report

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Summary

This report describes how vintages of the Real Time Database (RTD) on the ECB Data Portal can be built from 2001 to today, what we found when we checked the page, the files and the web service against each other, and what we suggest changing.

Vintages from January 2001 to December 2014 come as ready-made files. For later vintages the web service returns a log of changes, not a vintage. We rebuilt vintages from that log for all 278 series.

This report proceeds as

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1 Constructing the Vintages

We built the vintages in five steps. We found each series and its key in the ECB's list of series, downloaded the full log of changes for each one, and rebuilt a vintage by keeping, for every period, the latest row dated on or before the vintage's date. We then did this for all 278 series on each of the 261 release dates, and checked the result against the ECB's own vintages for 2001–2014 and against Eurostat. The steps below follow that order. The amber boxes mark the points where the page, the files or the web service could be read more than one way, and say how we read them.

Step 1: Finding a series and its key

1.1 Get the list of series. The page does not list the series the dataset holds. We found the list through the address below, which returns every series with its description.

<https://data-api.ecb.europa.eu/service/data/RTD?detail=nodata&format=csvdata>

The file holds 278 rows, one per series. Three of them, as an example, look like this.

KEY	TITLE	TITLE_COMPL (shortened)
RTD.Q.S0.S.G_GDPM_TO_C.E	Gross domestic product at market prices, volume	Gross domestic product at market price – Chain linked volumes – Euro
RTD.Q.S0.S.L_ULC_CONST.X	Unit Labour Cost (based on persons), deflator	Unit Labour Cost – Construction – Index
RTD.Q.S0.S.L_ULC_INDTO.X	Unit Labour Cost (based on persons), deflator	Unit Labour Cost – Industry, including energy – Index

1.2 Search by TITLE_COMPL, not TITLE. Different series can share a title. The two Unit Labour Cost rows above are different series with the same **TITLE**. **TITLE_COMPL** spells out what each series is. The file also gives each part of the key a column of its own, so we could filter the series by what they measure in the **RT_ECON_CONCEPT** column.

1.3 Read the key. Each series is named by a key of five parts, separated by dots. For real GDP, **Q.S0.S.G_GDPM_TO_C.E**, they are the frequency (Q, quarterly), the area (S0, the euro area), the seasonal adjustment (S, adjusted), the economic concept, meaning what the series measures (**G_GDPM_TO_C**, gross domestic product in chain-linked volumes), and the unit (E, euro). Where two rows share a **TITLE_COMPL**, they are the same series with and without seasonal adjustment, and the third part of the key tells them apart.

1.4 Take the key from the KEY column.

Ambiguity

The third part of the key does not always mean what it says. The ECB's code list defines **N** as "neither seasonally nor working day adjusted," but for M1, M2, M3, loans and credit the series with **N** holds the seasonally adjusted figures. For M3 growth it is the other way round. We found this only by comparing the series with the ECB's BSI dataset.

Step 2: Downloading its log

2.1 Drop RTD. from the key. The `KEY` column gives real GDP as `RTD.Q.Sø.S.G_GDPM_TO_C.E`, but the address already contains `RTD`, so only the part after it goes in. With `RTD.` left on the front of the key, the ECB refuses the request.

2.2 Request the series on its own, by its full five-part key. The address is

```
https://data-api.ecb.europa.eu/service/data/RTD/<key>?includeHistory=true&format=csvdata&detail=dataonly
```

and for real GDP

```
https://data-api.ecb.europa.eu/service/data/RTD/Q.Sø.S.G_GDPM_TO_C.E?includeHistory=true&format=csvdata&detail=dataonly
```

The option `includeHistory=true` asks for every version of the series rather than the latest one only. `format=csvdata` gives a CSV file, and `detail=dataonly` leaves out the columns of descriptive attributes, which we did not need to rebuild the vintages.

2.3 Download each series twice. We kept a file only if the two copies were identical and the last row was complete.

Ambiguity

Two behaviours of the web service look like success but are not. The address accepts a key with parts left empty, which fetches several series at once and is much faster. Such a request silently leaves out every removal, so periods the ECB withdrew would stay in the rebuilt vintages. That is why we requested each series with its full key. The server also sometimes sends a file cut short while reporting success, which is why we downloaded each series twice.

Step 3: Rebuilding a vintage from the log

What came back was not a vintage but a log of changes. The ECB updates the database on particular days, its releases. At each release it adds the newest periods and revises some earlier ones, and the log records each of these as a `Replace` row. Periods that had no revisions get no new row. Occasionally the ECB removes periods altogether, and the log records that as a `Delete` row. Each row gives the period it refers to (`TIME_PERIOD`), what happened to it (`ACTION`), the release on which a new value took effect (`VALID_FROM`) or on which the period was removed (`VALID_TO`), and the value itself (`OBS_VALUE`). We took a vintage to be the series as it stood on a given date, with the euro area's membership and the definitions in force at the time.

Ambiguity

Neither the page, the API help nor the technical documentation explains `ACTION`, `VALID_FROM` or `VALID_TO`. The names suggest that each value has a start and an end date, but a `Replace` row carries only `VALID_FROM` and a `Delete` row only `VALID_TO`. We worked out what they mean from the rows themselves.

To rebuild the vintage of a given date, we did the following for each period.

3.1 Keep only the rows dated on or before that date. Each row carries a single date, the day its new or revised value took effect (`VALID_FROM`) or the day its period was removed (`VALID_TO`).

3.2 Let the latest row kept decide the period's value.

- If it brings a new or revised value (**Replace**), that value (**OBS_VALUE**) is the period's value in the vintage.
- If it removes the period (**Delete**), the ECB had taken the period out by then, so the vintage has no value for it.
- If no row was kept, the ECB had not yet published the period, so the vintage has no value for it either.

3.3 Check the reading on an example. We used real GDP in the second half of 2005. These are the rows it holds between August and November 2005 for three of its quarters, in euro millions. They are the oldest quarter, 1991-Q1, which the ECB removed in that time, and the two newest, 2005-Q2 and 2005-Q3, which it added. The quarters in between are left out to keep the table short.

TIME_PERIOD	ACTION	VALID_FROM	VALID_TO	OBS_VALUE
1991-Q1	Replace	3 Aug 2005		1,275,507
1991-Q1	Replace	31 Aug 2005		1,273,486
1991-Q1	Replace	2 Nov 2005		1,275,571
1991-Q1	Delete		30 Nov 2005	
2005-Q2	Replace	31 Aug 2005		1,641,807
2005-Q2	Replace	2 Nov 2005		1,643,676
2005-Q2	Replace	30 Nov 2005		1,783,300
2005-Q3	Replace	30 Nov 2005		1,794,581

For the release of 30 November 2005, the latest row for 1991-Q1 is the **Delete** of that day, so the quarter is left out. For 2005-Q2 and 2005-Q3 it is the **Replace** of that day, so the vintage holds 1,783,300 and 1,794,581, the second appearing for the first time. Doing the same for each release gives the vintages.

Release date, 2005	3 Aug	31 Aug	5 Oct	2 Nov	30 Nov
1991-Q1	1,275,507	1,273,486	1,273,486	1,275,571	<i>removed</i>
2005-Q2	<i>not yet</i>	1,641,807	1,641,807	1,643,676	1,783,300
2005-Q3	<i>not yet</i>	<i>not yet</i>	<i>not yet</i>	<i>not yet</i>	1,794,581

Between them the two tables show every case. A quarter is added, as 2005-Q2 on 31 August. A quarter is revised, as 1991-Q1 on 31 August. A figure carries over because no row came in, as on 5 October. A quarter is removed, as 1991-Q1 on 30 November. The large revision to 2005-Q2 on 30 November was made on the same day the ECB removed 1991-1994.

Ambiguity

When a period carries both a **Delete** and a **Replace** on the same day, a rule that compares dates cannot choose between them, and letting the **Delete** win would make the series appear to lose data. We made the **Replace** win. Step 5 explains why this follows the ECB's own order.

Step 4: Building the whole dataset

We wanted the vintages to line up, so that every series has a vintage on every release and all 278 can be read as they stood on the same day.

4.1 Collect the release dates. The ECB publishes no list of the days on which it updated the database, but every date in a log's `VALID_FROM` or `VALID_TO` column is one. No single log holds them all, because a series only has a row on the releases where it changed. Pooling the dates in all 278 logs gave us the complete set, 261 releases between January 2001 and September 2026.

Ambiguity

The page says to take the vintage dates from the Governing Council's monetary policy decision dates, but the logs are stamped at 15:30 on the working day before. We dated each vintage by the stamp in the log, so the vintage for the meeting of 10 September 2026 is dated 9 September.

4.2 Rebuild every series at every release. We applied the rule from Step 3 to every series at each of the 261 dates. Every series gets a vintage on each release, carrying its figures over on the releases where it did not change.

4.3 Write the result in two forms. One is a workbook per series per vintage, 67,532 in all, in the layout used by ALFRED, the St. Louis Fed's archive of vintages, which suits reading a single vintage. The other is one table per series, with periods in rows and release dates in columns, which suits studying revisions.

The scripts, the results of the checks and the lists of out-of-date series are at <https://github.com/JChaquica/ecb-rtd-vintages>. The vintages themselves are too large to keep there and can be rebuilt with the scripts or supplied on request.

Step 5: Checking the result

We checked the vintages in four ways.

5.1 Against the ECB's own vintages for 2001–2014. The ECB published these as zip files, an independent record of what our reading of the log should produce. We rebuilt every series for each of the 168 months and compared it with that month's file. Where the file holds data for the series, 42,132 comparisons, the rebuilt vintage has the same periods and values in every one, with no tolerance for rounding. In a further 296 comparisons, covering 20 series, the file's column is empty but the rebuilt vintage is not. Most are the five government bond yields from February 2008 to March 2011, which the ECB stopped publishing after January 2008 but did not remove from the log until 6 April 2011.

Ambiguity

For these 296 comparisons the two ECB records disagree, and neither says which is right. We kept the log's values, since the method rests on the log, and `old_file_check.csv` lists every case for a user who wants to drop them.

5.2 Against Eurostat, for the later vintages. The ECB published no vintage files from 2015. Instead, we found that nine quarterly series equal Eurostat's figures for the euro area in every quarter since 1995, and that the newest quarter of real GDP carries the growth rate from the Eurostat release just before each vintage date.

5.3 Against the ECB's own order for same-day changes. When a period carries a removal and a

new value on the same day, the new value wins. This happens 34,989 times, all on 20 January and 8 September 2021, and in every case the removal is stamped one second before the new value, so the rule keeps the row the ECB entered last.

5.4 Against the files written. For every series we read back the latest workbook and one earlier one, 556 in all, and every date and value agreed with the rebuilt vintage, the values to the 15 significant digits Excel stores.

The zip files also hold 44 series that the web service does not have, most of them for the United States or Japan, and those can only be taken from the zip files. None of these checks shows that the ECB kept its series up to date.

2 Issues We Found

Below are the issues we faced while rebuilding the vintages and thought worth writing down.

Website documentation

Start of the history

Website states: Vintages from 2015 onwards can be extracted from the web service. **Found:** The service returns history to January 2001.

Update schedule

Website states: Updates are semi-annual, in January and July. **Found:** They occur before each Governing Council meeting: 12 times a year through 2014 and 8 times a year since.

Data before 2001

Website states: CSV vintages exist back to 1999 for 38 variables. **Found:** Only producer prices predate 2001, and the files are Excel.

Vintage reconstruction

Logs, not vintages

Each row is one change to one period, so for example, real GDP has 14,854 rows for 142 periods. A vintage is rebuilt by keeping, for each period, the latest row up to the cutoff.

Date-only cutoffs

Some revisions are a **Delete** at 15:30:00 and a **Replace** at 15:30:01 on the same day. Cutting by date alone can drop the period, so the full time stamp is needed.

Downloads

Missing history flag

Expected: A standard request returns the series history. **Found:** Without `includeHistory=true`, only today's values return, with no revision dates.

Wildcard series key

Expected: A wildcard matches a batch of individual requests. **Found:** The request succeeds but omits every **Delete** row, leaving removed periods in later vintages.

HTTP 200 status

Expected: A successful status means the response is complete. **Found:** A truncated file can still report success; the trailing **Delete** rows are lost.

Download speed

Full history takes 1–3 minutes per series and can time out. Because wildcards omit deletes, all 278 series must be requested separately.

Labels and dates

ACTION, VALID_FROM, VALID_TO

Label suggests: `VALID_FROM` and `VALID_TO` give each value a start and an end date. **Found:** A **Replace** has only `VALID_FROM`; a **Delete** has only `VALID_TO`. We could not find documentation on this.

Adjustment letter

Label suggests: `N` means neither seasonally nor working-day adjusted. **Found:** The code is reversed for ten money and credit series. Adjusted versions run through July 2026; the others stop at 4 June 2025.

Zip columns EZ and FG

Label suggests: `EZ` is M3 growth with key `N`; `FG` has key `Y`. **Found:** The reverse: `EZ` is seasonally adjusted and `FG` is unadjusted.

3 Data Status

Of the 278 series, 171 have a revision in 2026. The other 107 do not: 91 were last revised in 2025, 15 in 2024 and one in 2023. Nothing in the download marks them. The dates below are as the series stand in the vintage of 9 September 2026 at 15:30:01 Europe/Berlin. We took them from the raw history of each series, not from a label.

Two dates describe a series, and they answer different questions. *Last observation* is the latest period that still holds a value: a month for a monthly series, a quarter for a quarterly one, a year for an annual one. *Last revised* is the day the ECB last wrote a [Repl](#)ace row. A later last-revised date does not mean the series is up to date. EONIA was last revised on 10 April 2024, but its last observation is still December 2021.

3.1 Series that have stopped

A user who takes the latest vintage still sees numbers for these series, but they have not moved. Rows within each topic run from the oldest last revision to the newest.

Indicator	Series	Last observation	Last revised
National accounts, prices, and labour			
Taxes less subsidies on products	1	2024 Q4	16 Apr 2025
Government finance, annual, per cent of GDP	6	2024	4 Jun 2025
HICP excluding energy and unprocessed food	1	May 2025	4 Jun 2025
Labour cost index: sectors and seasonally adjusted total	4	2025 Q1	4 Jun 2025
Industry, retail, and trade			
New passenger-car registrations, key W	1	December 2022	15 Mar 2023
Retail trade, deflated: household equipment	1	December 2020	6 Mar 2024
Retail trade, deflated: textiles and clothing	1	April 2022	6 Mar 2024
Producer prices, non-durable consumer goods	1	October 2024	11 Dec 2024
Current account: goods, services, primary and secondary income	11	March 2025	4 Jun 2025
Export and import unit-value indices	13	February 2025	4 Jun 2025
Industrial production: including construction; total excluding construction and energy; durable and non-durable consumer goods	4	March 2025	4 Jun 2025
Producer prices, durable consumer goods	1	March 2025	4 Jun 2025
Retail trade, nominal total	1	March 2025	4 Jun 2025
Industrial production: total excluding construction (two keys), manufacturing, construction, energy, capital, intermediate and consumer goods	8	August 2025	29 Oct 2025
Producer prices, remaining components	7	August 2025	29 Oct 2025
Retail trade, deflated: total, food, and non-food	3	August 2025	29 Oct 2025
Surveys, money, and markets			
EONIA	1	December 2021	10 Apr 2024
Three-month Japanese yen Libor	1	January 2023	10 Apr 2024
Three-month US dollar Libor	1	September 2024	16 Oct 2024

Indicator	Series	Last observation	Last revised
Euro-area bond yields (five maturities), Brent crude oil, world raw-material prices (two)	8	November 2024	11 Dec 2024
Individual survey questions (order books, stocks, expectations)	15	May 2025	4 Jun 2025
M1, M2 and M3 stocks, loans, and loans and securities (key Y); M3 growth (key N); base money	7	April 2025	4 Jun 2025
United States and Japan			
Japan: unemployment rate	1	December 2023	6 Mar 2024
Japan and US GDP; US unit labour costs in manufacturing	3	2025 Q1	4 Jun 2025
Japan: manufacturing production, consumer prices, M2; US: manufacturing production, unemployment, M2	6	April 2025	4 Jun 2025
Total	107		

Eurostat's euro-area industrial production series runs through July 2026, while headline industrial production in the RTD ends in August 2025.

Three of these 107, EONIA and the three-month yen and dollar Libors, end because the underlying rate no longer exists.

3.2 The other 171 series

They include real, nominal and deflator GDP and their expenditure components (through 2026 Q2), total employment (2026 Q2), euro-area unemployment (July 2026), headline HICP (August 2026), the working-day adjusted total labour cost index (2026 Q2), goods trade values and volumes (May or June 2026), the government balances, compensation of employees and intermediate consumption (2025, last revised 29 April 2026), the capital account (June 2026), headline economic sentiment (August 2026), capacity utilisation (2026 Q3), the M1, M2 and M3 stocks, loans, and loans and securities with key N and M3 growth with key Y (July 2026), and US consumer prices, the S&P 500 and the Nikkei. GDP, total employment, euro-area unemployment and headline HICP were last revised on 9 September 2026.

3.3 Series that stalled, then caught up

These 46 series are current on 9 September 2026. Each sat still, or was empty, across a run of releases and then jumped. The columns are the last observation during that spell, the first release at which it appeared, and the release that added new periods. A user who opened a vintage in between would have seen the series as stopped.

A series counts as stalled here when, for at least two releases in a row, it was empty, its last observation was three months, two quarters or a year further behind than usual in 2015–2022, or it had gone longer than usual without any change. This is a judgement. A looser rule would add three quarterly series in 2022. Real GDP and capacity utilisation were one release late, and the total labour cost index two.

Group	Series	Last obs. while stuck	Unchanged from	Caught up	Now through
Residential construction prices	1	2021 Q3	2 Feb 2022	23 Jul 2025	2026 Q2
Effective exchange rates, monthly	5	Nov 2022	14 Dec 2022	23 Jul 2025	August 2026
Effective exchange rates, quarterly	1	2022 Q2	26 Oct 2022	23 Jul 2025	2026 Q1
Total employment and its components	14	2023 Q2	13 Sep 2023	23 Jul 2025	2026 Q2
Unemployment, by sex and age	10	Jan 2024	6 Mar 2024	23 Jul 2025	July 2026
Economic sentiment and the five confidence indicators	6	no data	1 Feb 2023	26 Jul 2023	August 2026
Capacity utilisation in manufacturing	1	2022 Q4	14 Dec 2022	26 Jul 2023	2026 Q3
Annual government series	8	2021	20 Jul 2022	26 Jul 2023	2025
Total	46				

For the eight annual government series a hold of about a year is normal, since each new year arrives once. What was late is the arrival. Figures for 2021 came on 20 July 2022 and figures for 2022 on 26 July 2023, in both years after the June release that usually brings them.

3.4 Empty or deleted

Nine series had no data in any vintage for years, because the log had removed every period, and then came back. The five bond yields and the two raw-material prices are empty from 6 April 2011 to 6 March 2024. Durable and non-durable producer prices are empty from 21 January 2015 to 6 March 2024. All nine return on 10 April 2024. They have since stopped again and sit in the table of 107.

Shorter gaps of the same kind also occurred. All 21 survey series were empty from 1 February to 14 June 2023 and returned on 26 July 2023. Six of them are current and appear in the table above, and the other 15 have since stopped. Four GDP deflators (domestic demand, government consumption, investment and private consumption) were empty on 5 November and 3 December 2014 and returned on 21 January 2015.